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# $\ell_1$-K-SVD: A robust dictionary learning algorithm with simultaneous update

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ABSTRACT

We develop a new dictionary learning algorithm called the $\ell_1$-K-SVD, by minimizing the $\ell_1$ distortion on the data term. The proposed formulation corresponds to maximum a posteriori estimation assuming a Laplacian prior on the coefficient matrix and additive noise, and is, in general, robust to non-Gaussian noise. The $\ell_1$ distortion is minimized by employing the *iteratively reweighted least-squares* algorithm. The dictionary atoms and the corresponding sparse coefficients are simultaneously estimated in the dictionary update step. Experimental results show that $\ell_1$-K-SVD results in noise-robustness, faster convergence, and higher atom recovery rate than the method of optimal directions, K-SVD, and the robust dictionary learning algorithm (RDL), in Gaussian as well as non-Gaussian noise. For a fixed value of sparsity, number of dictionary atoms, and data dimension, $\ell_1$-K-SVD outperforms K-SVD and RDL on small training sets. We also consider the generalized $\ell_p$, $0 < p < 1$, data metric to tackle heavy-tailed/impulsive noise. In an image denoising application, $\ell_1$-K-SVD was found to result in higher peak signal-to-noise ratio (PSNR) over K-SVD for Laplacian noise. The structural similarity index increases by 0.1 for low input PSNR, which is significant and demonstrates the efficacy of the proposed method.

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## 1. Introduction

Data-independent dictionaries such as discrete cosine transform (DCT), wavelet, curvelet, and contourlet have been extensively used for the applications of image denoising, compression, super-resolution, etc. Fixed dictionaries are generic, offer easy and efficient implementation, but are limited in their ability to provide a parsimonious representation of signals/images. The vast volume of research in compressive sensing [1,2], where one seeks to recover a sparse signal from its noisy linear measurements, has motivated researchers to intensify their quest for suitable dictionaries in which the signal of interest admits a sparse representation. When the dictionaries are *learnt* in a data-dependent or application specific manner, they lead to higher performance than their data-independent counterparts.

In this paper, we propose a dictionary learning approach that is robust to non-Gaussian noise contamination in the dataset, by applying a robust data fidelity metric, namely the $\ell_1$-norm. We demonstrate through experiments that the proposed approach not only results in robustness to additive noise, but is also more efficient than the competing algorithms when the size of the training dataset is small. Before proceeding with the developments, we provide a brief overview of some recent literature on dictionary learning.

### 1.1. Review of relevant literature

A survey of the state-of-the-art dictionary learning techniques is given in [3,4]. Seminal contributions to the problem of data-adaptive dictionary learning were made by Aharon et al. [5,6], who proposed the K-SVD algorithm,

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which, by far, is the most popular algorithm for data-dependent dictionary learning. Theoretical guarantees on the performance of K-SVD are available in [7,8]. Dai et al. [9] developed a generalized framework for dictionary learning, known as *simultaneous codeword optimization* (SimCO). The *method of optimal directions* (MOD) [10] and the K-SVD algorithms emerge as special cases of the generalized formulation.

K-SVD uses the $\ell_2$ distortion as a measure of data fidelity and finds applications in image denoising [5], where the noise is zero-mean, additive, and follows a white Gaussian distribution. The training is carried out either on the corpus of clean image patches or on patches derived from the noisy image itself. Since the K-SVD algorithm is not computationally efficient at handling large-dimensional image patches, a sparsity promoting prior is added locally on smaller patches, together with a global prior that enforces proximity between the noisy and the estimated denoised images.

Besides the synthesis model, the dictionary learning problem has also been addressed from the analysis and transform perspectives [11–14]. Apart from denoising, dictionary-based techniques also find applications in image super-resolution, inpainting, image restoration, etc. [15–17]. Discriminative sparse coding (DSC) algorithms such as discriminative K-SVD have been proposed in the literature for solving the signal classification problem [18,19]. Liu et al. [20] proposed the multiview Hessian discriminative sparse coding (MHDSC) algorithm, wherein one combines Hessian regularization with DSC for multiview classification problem. The MHDSC algorithm has been applied successfully for image annotation. Sun et al. [21] recently proposed an algorithm for learning discriminative dictionaries to solve the problem of image classification [22], where a combination of terms measuring discriminative fidelity, sub-dictionary incoherence, and group sparsity is minimized. Algorithms based on dictionary learning are not limited to only image processing applications. They have been widely used in other domains, including single channel speech separation [23], removal of ballistocardiogram effect from the EEG signals [24], human action recognition [25], to name a few. Dictionary learning is also equivalent to *soft clustering* and can be used to effectively learn inherently low-dimensional subspaces from large-dimensional data [26]. From a pattern classification perspective, dictionary learning can be interpreted as a technique to learn a union of low-dimensional subspaces from given training vectors [27].

In many applications, the assumption of Gaussianness of the additive noise may not be accurate, thus rendering $\ell_2$-minimization-based algorithms suboptimal. Moreover, $\ell_2$ minimization algorithms may lead to oversmoothing, causing loss of image details. One way to address the problem is to develop robust algorithms by minimizing the $\ell_1$ data metric. Algorithms for minimizing $\ell_1$-based data terms have been proposed in the context of non-negative matrix factorization (NMF) for document detection [28]. Liu and Tao have recently developed an NMF technique with the Manhattan or $\ell_1$ distortion metric [29], and carried out systematic analysis of the reconstruction error by decomposing it into approximation error and estimation error, with the latter bounded by the generalization error bound. They also determined rigorously how dimensionality reduction affects both errors. Recently, the problem of robust dictionary learning (RDL) has been addressed by Lu et al. [30] in an online setting, where the training examples are revealed sequentially [31]. Unlike K-SVD, the RDL algorithm does not offer the flexibility to simultaneously update the dictionary and the coefficients over the currently estimated support. A dictionary learning algorithm for impulse noise removal from images, through $\ell_1-\ell_1$ minimization, was proposed by Wang et al. [32], where an augmented Lagrangian framework is used to solve the resulting optimization problem. A notable contribution towards solving the regression problem in non-Gaussian noise was made by Liu and Tao [33], who addressed the regression problem in Cauchy distributed noise. They considered least-squares, least absolute deviation, and Cauchy loss, and showed that they are in the increasing order of robustness to outliers.

### 1.2. This paper

We develop a dictionary learning algorithm aimed at minimizing the $\ell_1$ distortion on the data term. The $\ell_1$ error is minimized using iteratively reweighted least-squares (IRLS) technique [34–36], employing which we solve the $\ell_1$ minimization problem by iteratively solving a series of $\ell_2$ minimization problems (cf. Section 2). The IRLS approach is appealing, because it enables one to minimize the $\ell_1$ cost in an iterative manner, where in each iteration, one essentially solves a weighted $\ell_2$ minimization problem, which has a closed-form solution. Consequently, the proposed method possesses the simplicity of the $\ell_2$ minimization problem, and the noise-robustness of the $\ell_1$ data penalty. The idea of reweighting is effective for developing computationally efficient algorithms for otherwise hard optimization problems. The notion of reweighting is also gaining importance in machine learning and pattern classification [37]. The use of $\ell_1$ metric for data fidelity results in robustness for suppression of heavy-tailed noise from images while preserving structure, measured in terms of structural similarity index (SSIM), as demonstrated by the experimental results (cf. Section 3). Unlike the RDL algorithm [30], in $\ell_1$-K-SVD, the dictionary atoms and the entries of the coefficient matrix are simultaneously updated (similar to K-SVD), offering better convergence performance. We show that the formulation also generalizes to the case of the $\ell_p$ metric, where $0<p<1$, which is known to promote sparsity better than the $\ell_1$ norm. The resulting $\ell_p$-K-SVD algorithm outperforms the K-SVD when the training data is corrupted by impulsive noise. Thus, the proposed method provides a unified framework to incorporate different penalties on the data term, customized based on the distribution of the additive noise.

## 2. Proposed dictionary learning approach

Denote the training dataset by $\mathcal{T}$, which contains $N$ exemplars $\{\mathbf{y}_n\}_{n=1}^{N}$ in $\mathbb{R}^m$, corrupted by additive noise. The objective is to learn a dictionary $D$ containing $K$ atoms, tailored to $\mathcal{T}$, such that $D$ represents the signals in $\mathcal{T}$ using sparse coefficient vectors $\mathbf{x}_n$. Typically, $D$ is overcomplete, that is, $m<K$. We use the symbols $Y$ and $X$ to denote the

matrices constructed by stacking the training exemplars $\mathbf{y}_n$ and the corresponding coefficients $\mathbf{x}_n$ in the columns, respectively. In order to achieve robustness to noise, we propose to solve the following optimization problem:

$$\min_{D,\mathbf{x}_n} \sum_{n=1}^{N} \left\|\mathbf{y}_n - D\mathbf{x}_n\right\|_1 + \sum_{n=1}^{N} \lambda_n \|\mathbf{x}_n\|_1 \quad \text{s.t. } \left\|\mathbf{d}_j\right\|_2 = 1 \text{ for all } j. \tag{1}$$

The unity norm constraint on the dictionary atoms helps avoid the ambiguity due to scale.

### 2.1. Interpretation and motivation for minimizing the $\ell_1$ objective

The cost function in (1) can be interpreted from a Bayesian perspective, where a Laplacian model is assumed for the prior on $\mathbf{x}_n$, as well as for the additive noise $\mathbf{w}_n$. The use of the $\ell_1$-metric on the data-term offers robustness to outliers, whereas its use as a regularizing penalty favors sparse $\mathbf{x}_n$. The parameters $\lambda_n$ trade off between sparsity and data fidelity. To explain further, assume that the observations follow the model $\mathbf{y}_n = D\mathbf{x}_n + \mathbf{w}_n$, where the zero-mean additive noise $\mathbf{w}_n$ is modeled using the Laplacian distribution

$$p_{\mathbf{w}}(\mathbf{w}_n) = \prod_{j=1}^{m} \frac{1}{2\gamma_n} \exp\left(-\frac{\left|(\mathbf{w}_n)_j\right|}{\gamma_n}\right) \quad \text{for all } n,$$

where each entry $(\mathbf{w}_n)_j$, $1 \le j \le m$, of the vector $\mathbf{w}_n$, is identically distributed with $\gamma_n$ as the variance parameter. Consider the Laplacian prior

$$p_{\mathbf{x}}(\mathbf{x}_n) = \prod_{j=1}^{K} \frac{1}{2\beta_n} \exp\left(-\frac{\left|(\mathbf{x}_n)_j\right|}{\beta_n}\right)$$

on the coefficient vectors $\mathbf{x}_n$. The maximum a posteriori (MAP) estimate of $\mathbf{x}_n$, which maximizes the log-posterior density $\log\left(p_{\mathbf{x}|\mathbf{y}}(\mathbf{x}_n|\mathbf{y}_n)\right)$, turns out to be

$$\arg\min_{\mathbf{x}_n} \left\|\mathbf{y}_n - D\mathbf{x}_n\right\|_1 + \frac{\gamma_n}{\beta_n}\|\mathbf{x}_n\|_1. \tag{2}$$

Denoting $\lambda_n = \frac{\gamma_n}{\beta_n}$, introducing $D$ as a variable to be optimized over, and summing over all observations, leads to the optimization posed in (1).

### 2.2. Development of the $\ell_1$-K-SVD algorithm

The overall cost-function in (1) is convex in $D$ for a fixed $X$, and vice versa. However, it is not jointly convex in $(D,X)$. Therefore, to solve (1), we adopt an alternating minimization strategy, wherein one starts with an initial guess for $D$, and updates $\mathbf{x}_n$ and $D$, alternately.

#### 2.2.1. Sparse coding

To update the coefficient vectors $\mathbf{x}_n$ when $D$ is fixed, one needs to solve $N$ independent minimization problems of the form

$$\min_{\mathbf{x}_n} \left\|\mathbf{y}_n - \hat{D}\mathbf{x}_n\right\|_1 + \lambda_n \|\mathbf{x}_n\|_1, \tag{3}$$

where $\hat{D}$ denotes the current estimate of the dictionary. To solve (3) using IRLS, one starts with an initial guess $\mathbf{x}_n^{(0)}$ and updates $\mathbf{x}_n$ in the $k^{\text{th}}$ iteration as

$$\mathbf{x}_n^{(k+1)} = \left(\hat{D}^T W_{1n}^{(k)} \hat{D} + \lambda_n W_{2n}^{(k)}\right)^{-1} \hat{D}^T W_{1n}^{(k)} \mathbf{y}_n,$$

where $W_{1n}$ and $W_{2n}$ are diagonal weight matrices of appropriate dimensions. Typically, one begins the iterations by setting $W_{1n}^{(0)}$ and $W_{2n}^{(0)}$ to identity matrices, for all $n$, and updates them in the $k^{\text{th}}$ iteration as

$$W_{1n}^{(k+1)}(j) = \frac{1}{\left|\left(\mathbf{y}_n - \hat{D}\mathbf{x}_n^{(k)}\right)_j\right| + \epsilon} \quad \text{and} \quad W_{2n}^{(k+1)}(j) = \frac{1}{\left|\left(\mathbf{x}_n^{(k)}\right)_j\right| + \epsilon}, \tag{4}$$

where $W(j)$ denotes the $j^{\text{th}}$ entry of the diagonal matrix $W$, and $(\mathbf{x})_j$ denotes the $j^{\text{th}}$ entry of the vector $\mathbf{x}$. A small positive constant $\epsilon$ is added to the denominator to ensure numerical stability of the algorithm. One can also consider an equivalent constrained version of (3) of the form

$$\min_{\mathbf{x}_n} \left\|\mathbf{y}_n - \hat{D}\mathbf{x}_n\right\|_1 \quad \text{s.t. } \|\mathbf{x}_n\|_1 \le \tau_n, \tag{5}$$

where $\tau_n$ controls the sparsity of the resulting solution, with smaller values of $\tau_n$ resulting in higher sparsity. Efficient optimization packages such as CVX [38] and ManOpt [39] are available to solve problems of the form given in (3) and (5). To reduce noise sensitivity, we apply a threshold $T_0$ on the entries of $X$, after sparse coding. The optimum threshold value is application-specific, and the values chosen in our experiments are indicated in Section 3. Thus, the proposed approach may be interpreted as $\ell_1$-minimization followed by pruning.

#### 2.2.2. Dictionary update

In the RDL algorithm [30], one updates the dictionary by solving

$$\min_{D} \left\|Y - D\hat{X}\right\|_1 \quad \text{s.t. } \left\|\mathbf{d}_j\right\|_2 = 1 \text{ for all } j,$$

for a fixed sparse coefficient matrix $\hat{X}$. The symbol $\|A\|_1$ denotes the sum of the absolute values of the entries of matrix $A$. This approach may be viewed as the $\ell_1$-counterpart of the MOD algorithm, which does not offer the flexibility to update $X$ over its currently estimated support, along with the atoms of the dictionary $D$. In $\ell_1$-K-SVD, one fixes the support of $X$ at its current estimate $\Omega$, and updates $D$ and $X$ over the support set $\Omega$, simultaneously, by solving

$$\min_{D,X} \|Y - DX\|_1 \quad \text{s.t. } \left\|\mathbf{d}_j\right\|_2 = 1 \text{ for all } j \text{ and } \operatorname{supp}(X) \subseteq \Omega,$$

where $\operatorname{supp}(X)$ denotes the set of non-zero entries of the matrix $X$.

#### 2.2.3. Theoretical explanation and algorithm for simultaneous update

Denote $J(D,X) = \|Y - DX\|_1$, which is the objective function to be minimized. We denote the current estimate of the coefficient matrix $X$ and its support, prior to dictionary update, by $\hat{X}$ and $\Omega$, respectively. To analyze the reduction in the cost function using the simultaneous

update strategy, we define

$$\hat{X}_{wsu} = \hat{X}, \tag{6a}$$

$$\hat{D}_{wsu} = \arg\min_{D} J\left(D, \hat{X}\right) \quad \text{s.t. } \|\mathbf{d}_j\|_2 = 1 \text{ for all } j \quad \text{and} \tag{6b}$$

$$\left(\hat{D}_{su}, \hat{X}_{su}\right) = \arg\min_{D,X} J(D,X) \quad \text{s.t. } \|\mathbf{d}_j\|_2 = 1 \text{ for all } j, \quad \text{supp}(X) \subseteq \Omega, \tag{6c}$$

where the subscripts *wsu* and *su* denote that the estimates are obtained without and with a simultaneous updating strategy, respectively. From (6c), it follows that

$$J\left(\hat{D}_{su}, \hat{X}_{su}\right) \le J(D,X),$$

for any $D \in \mathcal{D}$ and any $X \in \mathcal{X}_\Omega$, where $\mathcal{D} = \{D: \|\mathbf{d}_j\|_2 = 1 \text{ for all } j\}$ and $\mathcal{X}_\Omega = \{X: \text{supp}(X) \subseteq \Omega\}$. In particular, $\hat{D}_{wsu} \in \mathcal{D}$ by (6b) and $\hat{X}_{wsu} = \hat{X} \in \mathcal{X}_\Omega$ by hypothesis, since the support of $\hat{X}$ is defined to be $\Omega$. Therefore, we have that $J\left(\hat{D}_{su}, \hat{X}_{su}\right) \le J\left(\hat{D}_{wsu}, \hat{X}_{wsu}\right)$, which indicates that the simultaneous updating strategy results in larger reduction in the value of the objective as compared with the case where the coefficient matrix $X$ is kept fixed. Empirically, such a joint update strategy was found to result in better convergence, as indicated by the experimental results in Section 3.1.

To simultaneously update the $j^{\text{th}}$ atom in the dictionary and the corresponding row in $X$, we focus on the error matrix $E_j = Y - \sum_{i \ne j} \mathbf{d}_i \mathbf{x}^i$, resulting from the absence of $\mathbf{d}_j$, where $\mathbf{x}^i$ denotes the $i^{\text{th}}$ row of $X$. In order to obtain a refined estimate of $\mathbf{d}_j$ and $\mathbf{x}^j$, one is required to solve the optimization problem:

$$\min_{\mathbf{d}_j, \mathbf{x}^j} \|E_j - \mathbf{d}_j \mathbf{x}^j\|_1 \quad \text{s.t. } \mathcal{S}\left(\mathbf{x}^j\right) = \Omega_j \quad \text{and} \quad \|\mathbf{d}_j\|_2 = 1. \tag{7}$$

In (7), we seek to minimize the cost function subject to the constraint that $\mathcal{S}(\mathbf{x}^j) = \Omega_j$, where $\mathcal{S}(\cdot)$ denotes the support operator for a vector, defined as $\mathcal{S}(\mathbf{z}) = \{i: \mathbf{z}_i \ne 0\}$. Therefore, it suffices to consider only those columns of $E_j$ whose indices are in the support set $\Omega_j$, denoted by $E_j|_{\Omega_j}$. Furthermore, to get rid of scale ambiguity, we restrict $D$ to have unit-length atoms, that is, $\|\mathbf{d}_j\|_2 = 1$ for all $j$. Consequently, (7) is equivalent to

$$\min_{\mathbf{u},\mathbf{v}} \|E - \mathbf{u}\mathbf{v}^T\|_1 = \min_{\mathbf{u},\mathbf{v}} \sum_{n=1}^{M} \|\mathbf{e}_n - v_n \mathbf{u}\|_1 \quad \text{s.t. } \|\mathbf{u}\|_2 = 1, \tag{8}$$

where $\mathbf{e}_n$ is the $n^{\text{th}}$ column of the matrix $E$ and $v_n$ denotes the $n^{\text{th}}$ entry of the vector $\mathbf{v}$. The key idea behind the $\ell_1$-K-SVD approach is to approximate the $\ell_1$-norm using a reweighted $\ell_2$ metric:

$$\sum_{n=1}^{M} \|\mathbf{e}_n - v_n \mathbf{u}\|_1 \approx \sum_{n=1}^{M} (\mathbf{e}_n - v_n \mathbf{u})^T W_n (\mathbf{e}_n - v_n \mathbf{u}).$$

Differentiating with respect to $\mathbf{u}$ and setting to zero, for a fixed $v$, we get that

$$\mathbf{u} = \left(\sum_{n=1}^{M} v_n^2 W_n\right)^{-1} \left(\sum_{n=1}^{M} v_n W_n \mathbf{e}_n\right).$$

Similarly, for a fixed $\mathbf{u}$, we have $v_n = \frac{\mathbf{u}^T W_n \mathbf{e}_n}{\mathbf{u}^T W_n \mathbf{u}}$. Once the update of $\mathbf{u}$ and $\mathbf{v}$ is done, the $j^{\text{th}}$ diagonal element of $W_n$ should be updated as $W_n(j) \leftarrow \frac{1}{|(\mathbf{e}_n - v_n \mathbf{u})_j| + \epsilon}$. Dictionary update is performed by setting $\mathbf{d}_j = \mathbf{u}_0$ and $\mathbf{x}^j|_{\Omega_j} = \mathbf{v}_0^T$, where $(\mathbf{u}_0, \mathbf{v}_0)$ solves (8). The step-by-step description to obtain the solution to (8) is given in Algorithm 1. We chose $J = 10$ in our experiments. The $\ell_1$-K-SVD algorithm is computationally more expensive than the K-SVD and starts with the K-SVD initialization. However, the quality of the learnt dictionary, measured in terms of its proximity to the ground truth, is better for $\ell_1$-K-SVD than K-SVD, for both non-Gaussian and Gaussian noise conditions.

**Algorithm 1.** ($\ell_1$-**K-SVD**): To find $\mathbf{u}$ and $\mathbf{v}$ such that $\|E - \mathbf{u}\mathbf{v}^T\|_1$ is minimized subject to $\|\mathbf{u}\|_2 = 1$.

1. **Input:** Error matrix $E$, number of iterations $J$.
2. **Initialization:**
   - Set the iteration counter $t \leftarrow 0$, $\mathbf{u}^{(t)} \leftarrow \mathbf{a}$, and $\mathbf{v}^{(t)} \leftarrow \sigma \mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are the left and right singular vectors, respectively, corresponding to the largest singular value $\sigma$ of $E$.
   - Initialize the weight matrices $W_n^{(t)} \leftarrow I$, for $n = 1, 2, \ldots, M$, where $I$ denotes the identity matrix and $M$ is the number of columns in $E$.
3. **Iterate the following steps $J$ times:**
   1. $W_n^{(t+1)}(j) \leftarrow \frac{1}{\left|(\mathbf{e}_n - v_n^{(t)} \mathbf{u}^{(t)})_j\right| + \epsilon}$
   2. $\mathbf{u}^{(t+1)} \leftarrow \left[\sum_{n=1}^{M} \left(v_n^{(t)}\right)^2 W_n^{(t+1)}\right]^{-1} \left[\sum_{n=1}^{M} v_n^{(t)} W_n^{(t+1)} \mathbf{e}_n\right]$
   3. $v_n^{(t+1)} \leftarrow \frac{(\mathbf{u}^{(t+1)})^T W_n \mathbf{e}_n}{(\mathbf{u}^{(t+1)})^T W_n (\mathbf{u}^{(t+1)})}$, for $n = 1, 2, \ldots, M$
   4. $t \leftarrow t + 1$
4. **Scaling:** Set $\mathbf{u} \leftarrow \frac{\mathbf{u}^{(t)}}{\|\mathbf{u}^{(t)}\|_2}$ and $\mathbf{v} \leftarrow \|\mathbf{u}^{(t)}\|_2 \mathbf{v}^{(t)}$
5. **Output:** Solution $(\mathbf{u}, \mathbf{v})$ to (8).

### 2.3. The $\ell_p$-K-SVD algorithm

The $\ell_1$-K-SVD algorithm can be generalized to handle sparse/impulsive noise, where the measurement vectors are corrupted only at a few locations. Since $\ell_p$-metrics, for $0 < p < 1$, promote sparsity better than the $\ell_1$-metric, we propose to solve the $\ell_p$-regularized $\ell_p$-minimization program, given by

$$\min_{D, \mathbf{x}_n} \sum_{n=1}^{N} \|\mathbf{y}_n - D\mathbf{x}_n\|_p^p + \sum_{n=1}^{N} \lambda_n \|\mathbf{x}_n\|_p^p. \tag{9}$$

The essential idea behind this approach remains the same as $\ell_1$-K-SVD, in the sense that we use a quadratic approximation to the $\ell_p$-metric and minimize that iteratively, with appropriately updated weight matrices.

The cost function in (9) corresponds to a Bayesian formulation if one assumes a generalized $p$-Gaussian distribution of the form

$$p(\mathbf{u}) = \prod_j \frac{p}{2\Gamma(1/p)\gamma\sigma_e} \exp\left(-\frac{|(\mathbf{u})_j|^p}{(\gamma\sigma_e)^p}\right), \tag{10}$$

where $\gamma = \left(\frac{\Gamma(1/p)}{\Gamma(3/p)}\right)^{1/2}$, on $\mathbf{x}_n$'s and $\mathbf{w}_n$'s, albeit with different values of the parameter $\sigma_e$. For $0 < p \le 1$, samples drawn from the distribution in (10) yield spiky sequences [40]. Consequently, the MAP formulation, which is equivalent to minimizing the $\ell_p$ metric for the statistical model in (10), results in a sparse solution. The IRLS algorithm, which is appropriately modified for the $\ell_p$-norm penalty as explained next, is employed to solve (9).

In the sparse coding stage, instead of using (4), one should update the weight matrices following the update

rules

$$W_{1n}^{(k+1)}(j)=\frac{1}{\left|\left(\mathbf{y}_n-\hat{D}\mathbf{x}_n^{(k)}\right)_j\right|^{2-p}+\epsilon} \quad \text{and} \quad W_{2n}^{(k+1)}(j)=\frac{1}{\left|\left(\mathbf{x}_n^{(k)}\right)_j\right|^{2-p}+\epsilon}. \tag{11}$$

For dictionary update in case of the $\ell_p$-metric, one is required to solve an optimization problem of the form

$$\min_{\mathbf{u},\mathbf{v}}\|E-\mathbf{u}\mathbf{v}^T\|_p^p=\min_{\mathbf{u},\mathbf{v}}\sum_{n=1}^{M}\|\mathbf{e}_n-v_n\mathbf{u}\|_p^p \quad \text{s.t. } \|\mathbf{u}\|_2=1, \tag{12}$$

where $\|A\|_p^p \triangleq \sum_{i,j}|A_{ij}|^p$, for a matrix $A$. In order to solve (12), Step 3.1 in Algorithm 1 is modified as

$$W_n^{(t+1)}(j)\leftarrow\frac{1}{\left|\left(\mathbf{e}_n-v_n^{(t)}\mathbf{u}^{(t)}\right)_j\right|^{2-p}+\epsilon}.$$

As in the case of $\ell_1$-K-SVD, a positive constant $\epsilon$ is added to the denominator to ensure numerical stability by avoiding divide-by-zero cases. We refer to the resulting algorithm as the $\ell_p$-K-SVD algorithm. A comparison of $\ell_p$-K-SVD with K-SVD under sparse noise corruption is given in Section 3.2.

## 3. Experimental results

To validate the performance of the proposed algorithms, we carry out experiments on synthesized signals as well as real images. The performance of the $\ell_1$-K-SVD algorithm is compared with the K-SVD, MOD, and the RDL algorithms. The MOD and K-SVD algorithms use the $\ell_2$ data term, whereas the RDL algorithm minimizes the $\ell_1$ error on the data term. Unlike the MOD and the RDL algorithms, the K-SVD algorithm is equipped with a simultaneous updating strategy during the dictionary estimation step. The proposed $\ell_1$-K-SVD algorithm is designed to minimize $\ell_1$ data error to ensure robustness to noise and equipped with simultaneous update to yield faster convergence. The details of the setup and findings of the experiments are given next.

### 3.1. on Performance of $\ell_1$-K-SVD synthesized signal

To validate the $\ell_1$-K-SVD algorithm, we first conduct experiments on synthesized training data, for Gaussian as well as Laplacian noise. Since the ground truth dictionary $D$ is known in the experiments, we use two performance metrics, namely the atom detection rate (ADR) and distance $\kappa\left(\hat{D},D\right)$ of the estimated dictionary $\hat{D}$ from the ground truth $D$. To compute ADR, we count the number of recovered atoms in $D$. An atom $\mathbf{d}_i$ in $D$ is considered to have been recovered if $\left|\mathbf{d}_i^T\hat{\mathbf{d}}_j\right|$ exceeds 0.99 for some $j$, $1\le j\le K$. Since $D$ and $\hat{D}$ contain unit-length atoms, this criterion ensures a near-accurate atom recovery. Finally, to compute ADR, we calculate the ratio of the number of recovered atoms to the total number of atoms $K$. The metric $\kappa$ measures the closeness of $\hat{D}$ to $D$, and is defined as

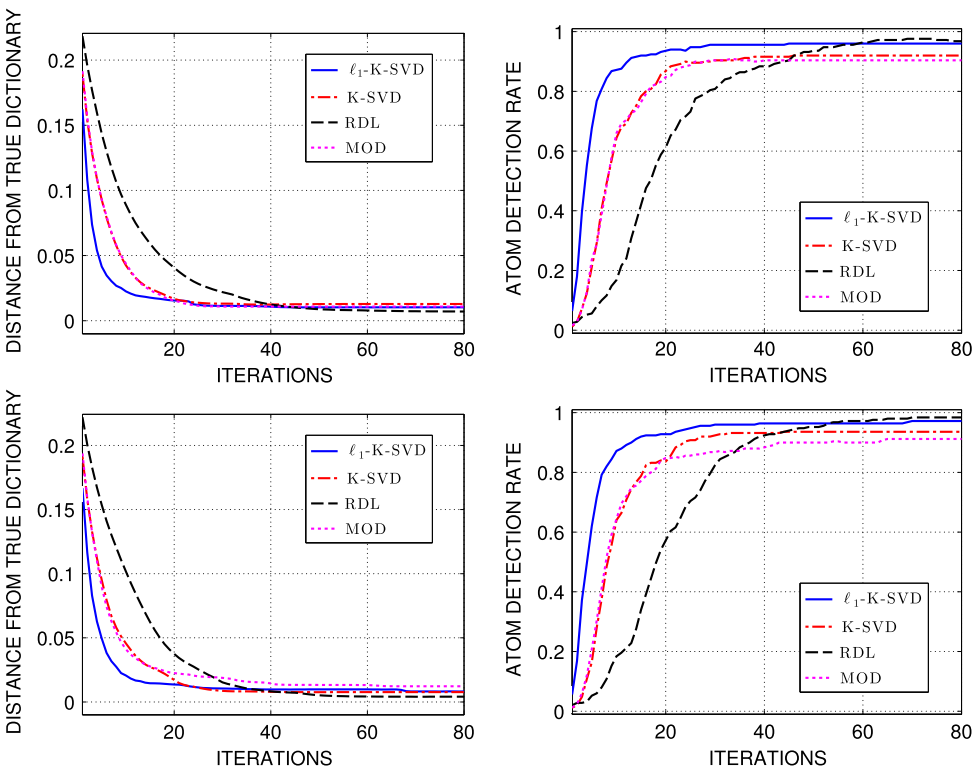


**Fig. 1.** Performance comparison of the $\ell_1$-K-SVD, K-SVD, RDL, and MOD algorithms in terms of atom detection rate (ADR) and distance metric $\kappa$, for Gaussian (first row) and Laplacian (second row) noise. The plots are averaged over five independent random trials. The measurement SNR is 20 dB. The number of examples in the training set is $N=1500$. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

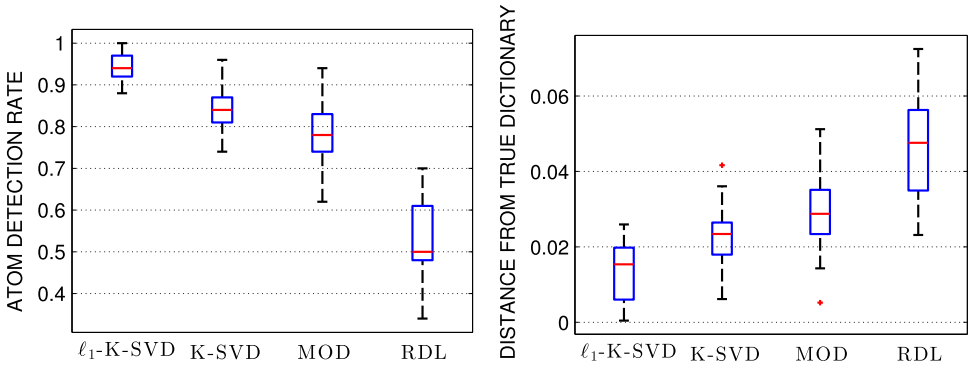


**Fig. 2.** Performance comparison of different algorithms in case of Laplacian noise contamination with an input SNR level of 20 dB. The variations in the atom detection rate (ADR) and the distance $\kappa$ of the recovered dictionary from the ground truth after 20 iterations are shown. The variations are computed over 20 independent random trials. The $\ell_1$-K-SVD algorithm exhibits a larger median value for the ADR and a relatively smaller median value for $\kappa$ over the competing algorithms. The variance about the median value is found to be smaller for the $\ell_1$-K-SVD algorithm. (For interpretation of the references to color in this figure, the reader is referred to the web version of this paper.)

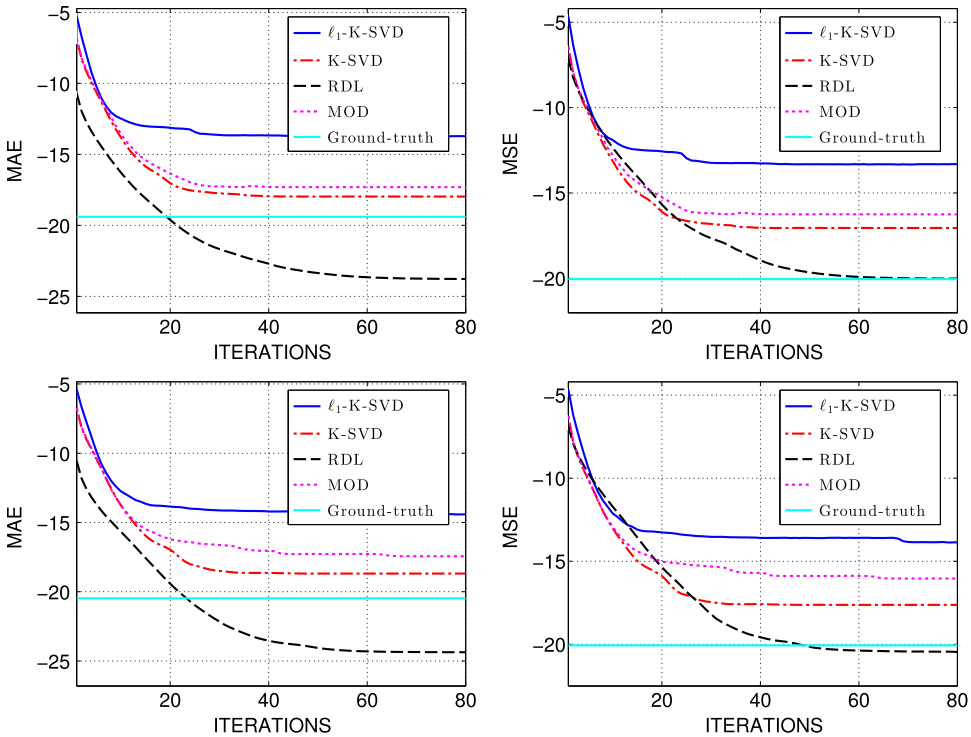


**Fig. 3.** The MAE and MSE for different algorithms on logarithmic scale. The first and second rows correspond to the additive Gaussian and Laplacian noise contaminations, respectively. The horizontal magenta line in each plot corresponds to the $\ell_2$ norm of the noise in case of the MSE, and the $\ell_1$ norm in case of the MAE. The measurement SNR is 20 dB. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

$\kappa = \frac{1}{K}\sum_{i=1}^{K} \min_{1 \le j \le K}\left(1 - \left|\mathbf{d}_i^T \hat{\mathbf{d}}_j\right|\right)$. The value of $\kappa$ satisfies $0 < \kappa < 1$. If $\kappa$ is close to 0, it indicates that the recovered dictionary closely matches with the ground truth. We also compare the evolution of the data representation error of the competing algorithms, measured in terms of the mean absolute error (MAE) and mean-squared error (MSE), defined as

$$\text{MAE} = \frac{\|Y - DX\|_1}{\|Y\|_1} \quad \text{and} \quad \text{MSE} = \frac{\|Y - DX\|_F^2}{\|Y\|_F^2},$$

respectively, as a function of the number of iterations. These two metrics indicate how well the recovered dictionaries explain the training dataset $Y$.

The experimental setup is identical to that in [6]. The ground truth dictionary is created by randomly generating $K = 50$ vectors in $\mathbb{R}^{20}$, followed by column normalization. Each column of the coefficient matrix $X$ has $s = 3$ nonzeros, with their locations chosen uniformly at random and amplitudes drawn from the $\mathcal{N}(0, 1)$ distribution. Subsequently, $D$ is combined with $X$ and contaminated by additive noise, to generate $N = 1500$ exemplars. All algorithms

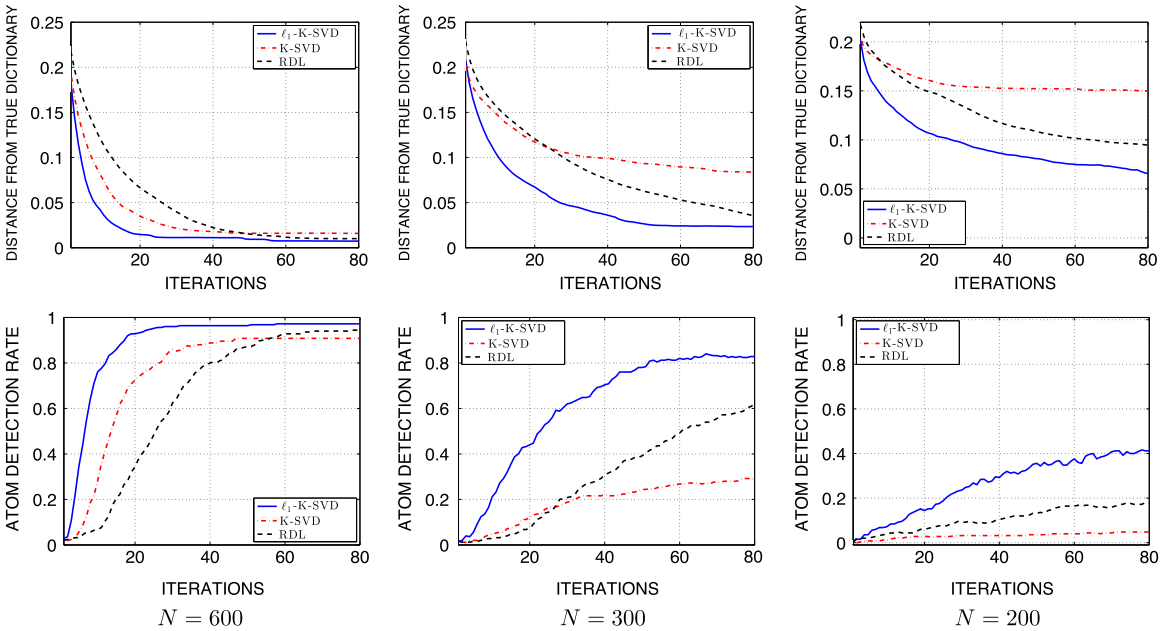


**Fig. 4.** Performance comparison of $\ell_1$-K-SVD, K-SVD, and RDL algorithms for varying data-size $N$. The values of $m$, $s$, and $K$ are kept fixed at $m=20$, $s=3$, and $K=50$. The figures in the first and second rows show the variations of the distance metric $\kappa$ and the ADR, respectively, with iterations. The training dataset is corrupted by additive Laplacian noise with measurement SNR of 20 dB. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

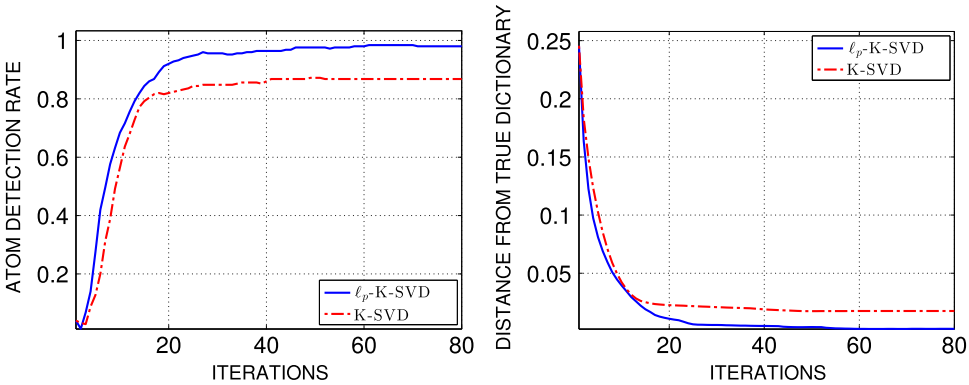


**Fig. 5.** Performance comparison of $\ell_p$-K-SVD and K-SVD for sparse noise contamination, in terms of ADR and the distance metric $\kappa$. We have used the $\ell_p$-metric for the data-fidelity term as well as the regularizer, where $p=\frac{1}{2}$. The measurement SNR is 30 dB. The plots are obtained by averaging over five independent random trials. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

are initialized with the training examples and the iterations are repeated 80 times, as suggested in [6]. The optimum choice of the hard-threshold $T_0$, determined experimentally, is given by $T_0 = 0.03\|X\|_F$, where $\|\cdot\|_F$ denotes the Frobenius norm. The variation of $\kappa$ and ADR with iterations, averaged over five independent trials, is shown in Fig. 1. To facilitate a fair comparison of $\ell_1$-K-SVD with the K-SVD, MOD, and RDL [30] algorithms, true values of the $\ell_1$ and $\ell_0$ norms of the ground truth coefficient vectors, that is, true values of $\tau_n$ in (5) and the sparsity parameter $s$, respectively, are supplied to the algorithms. Orthogonal matching pursuit (OMP) is used for sparse coding in K-SVD and MOD.

From Fig. 1, we observe that the $\ell_1$-K-SVD algorithm results in superior performance over RDL and its $\ell_2$ counterparts, namely K-SVD and MOD, both in terms of $\kappa$ and ADR, for both Gaussian and Laplacian noise conditions. The faster convergence of the $\ell_1$-K-SVD algorithm compared to RDL is attributed to the simultaneous update of the dictionary and the coefficients. However, they result in similar performance after sufficiently many iterations. The execution times taken for each iteration of the K-SVD, MOD, RDL and $\ell_1$-K-SVD algorithms are 0.53, 0.48, 45.53, and 32.90 s, respectively, when executed on a MATLAB 2011 platform, running on a Macintosh OSX system with 8 GB RAM and 3.2 GHz core-i5 processor.

To analyze the performance variation in the algorithms, we compute the ADR and the $\kappa$ metric for the competing algorithms over 20 independent random simulations, after 20 iterations in each simulation. The input training set is contaminated by additive Laplacian noise, with an input SNR of 20 dB. The median values of ADR and $\kappa$, indicated by the central red marks, are shown in Fig. 2. The blue

boxes indicate the 25 and 75 percentile points, whereas the extreme black bars denote the maximum and minimum values. We observe that the $\ell_1$-K-SVD algorithm results in a relatively smaller value for $\kappa$ and larger value for the ADR metric, as compared with the K-SVD, MOD, and RDL algorithms. It also has smaller variation around the median. This experiment illustrates the robustness, reliability, and stability of the $\ell_1$-K-SVD algorithm in recovering the underlying ground truth dictionary.

The data-fitting error, measured in terms of the MAE and MSE, is plotted in Fig. 3, for additive Laplacian and Gaussian noise contaminations. We observe that $\ell_1$-K-SVD yields a slower rate of decay for MAE and MSE in comparison with MOD, K-SVD, and RDL algorithms. The MAE obtained using the RDL algorithm goes below the ground truth value of MAE, which is equal to $\|W\|_1$, where $W$ denotes the additive noise. This experiment indicates that RDL tends to over-fit the training data, leading to inferior performance when the dictionary learnt is applied to unseen signals. The reason for the relatively higher data-fitting error for $\ell_1$-K-SVD lies in the thresholding step, following the sparse coding operation. The choice of the threshold $T_0$ trades off between the data-fitting error and the noise rejection capability of the algorithm. Choosing a higher value of $T_0$ leads to better noise suppression, at the cost of poorer data-fitting.

**Table 1**
Values of the regularization parameter $\lambda$ and the fraction of entries $n_p$ retained in the thresholding step of $\ell_1$-K-SVD, in case of Laplacian (black) and Gaussian (red) noise, for various values of $\sigma$.

| $\sigma = 15$ | $\sigma = 25$ | $\sigma = 35$ |
|---|---|---|
| $\lambda = 1, 1$ | $\lambda = 1, 1$ | $\lambda = 8, 8$ |
| $n_p = 0.18, 0.15$ | $n_p = 0.08, 0.05$ | $n_p = 0.05, 0.04$ |

### 3.1.1. *Effect of the size of training dataset*

To analyze the effect of the training data size on the performance of the algorithms, we decrease the value of $N$, keeping $m$, $s$, and $K$ fixed. From the plots in Fig. 4, we observe that the $\ell_1$-K-SVD algorithm is superior to RDL and K-SVD, when the value of $N$ is small. For $N=200$, we note that the $\ell_1$-K-SVD algorithm recovers approximately 40% of the atoms accurately, whereas the ADR for RDL and K-SVD is below 20%. This experiment suggests that the $\ell_1$-K-SVD is particularly suitable when the number of training examples available for learning the dictionary is small.

## 3.2. *Performance of $\ell_p$-K-SVD*

Experiments are conducted on synthesized signals corrupted by sparse noise, to demonstrate the efficiency of the $\ell_p$-K-SVD algorithm over K-SVD. All experimental settings, other than the noise generation process, are kept unaltered as in Section 3.1; $\lambda_n$ in (9) are all set to unity. The additive noise is generated by creating a sparse matrix, having two nonzero entries in every column, whose locations are selected uniformly at random from $m$ possibilities and the magnitudes are drawn from the distribution $\mathcal{N}(0,1)$. The nonzero entries of the noise matrix are appropriately scaled to achieve a requisite measurement SNR, defined as

$$\mathrm{SNR} = 20\log_{10}\left(\frac{\|Y_0\|_F}{\|W\|_F}\right)\ \mathrm{dB},$$

where $Y_0$ denotes the uncorrupted measurement matrix and $W$ is the additive noise. The measurement SNR in the experiment is chosen to be 30 dB. Following the sparse coding step, a threshold $T_0 = 0.03\|X\|_F$ is applied to the entries of $X$, to prune the entries having small magnitudes. The factor 0.03 in the expression of $T_0$ is observed to

**Table 2**
Comparison of output PSNR (dB) and SSIM (corresponding to the first and second rows in each cell, respectively) for the K-SVD and $\ell_1$-K-SVD algorithms on the *House* and *Tower* images corrupted by additive Laplacian noise, for different input noise levels. The values are obtained by averaging over five indpendent realizations of noise. The values of PSNR and SSIM do not change much from one realization to another.

| Images | K-SVD | | | $\ell_1$-K-SVD | | |
|---|---|---|---|---|---|---|
| | $\sigma = 15$ | $\sigma = 25$ | $\sigma = 35$ | $\sigma = 15$ | $\sigma = 25$ | $\sigma = 35$ |
| | 31.50 | 28.39 | 26.12 | 31.70 | 28.46 | 26.79 |
| | 0.79 | 0.67 | 0.55 | 0.81 | 0.76 | 0.66 |
| | 31.04 | 27.98 | 25.86 | 30.96 | 28.30 | 26.55 |
| | 0.80 | 0.66 | 0.53 | 0.80 | 0.75 | 0.60 |

**Table 3**
Comparison of output PSNR (dB) and SSIM (corresponding to the first and second rows in each cell, respectively) for the K-SVD and $\ell_1$-K-SVD algorithms on the *House* and *Tower* images corrupted by additive white Gaussian noise for different input noise levels. The values are averaged over five independent noise realizations.

| Images | K-SVD | | | $\ell_1$-K-SVD | | |
|---|---|---|---|---|---|---|
| | $\sigma = 15$ | $\sigma = 25$ | $\sigma = 35$ | $\sigma = 15$ | $\sigma = 25$ | $\sigma = 35$ |
| | 31.72 | 28.66 | 26.43 | 30.77 | 28.58 | 25.98 |
| | 0.80 | 0.68 | 0.56 | 0.78 | 0.73 | 0.62 |
| | 31.23 | 28.26 | 26.18 | 30.08 | 28.00 | 26.22 |
| | 0.81 | 0.67 | 0.54 | 0.77 | 0.73 | 0.61 |

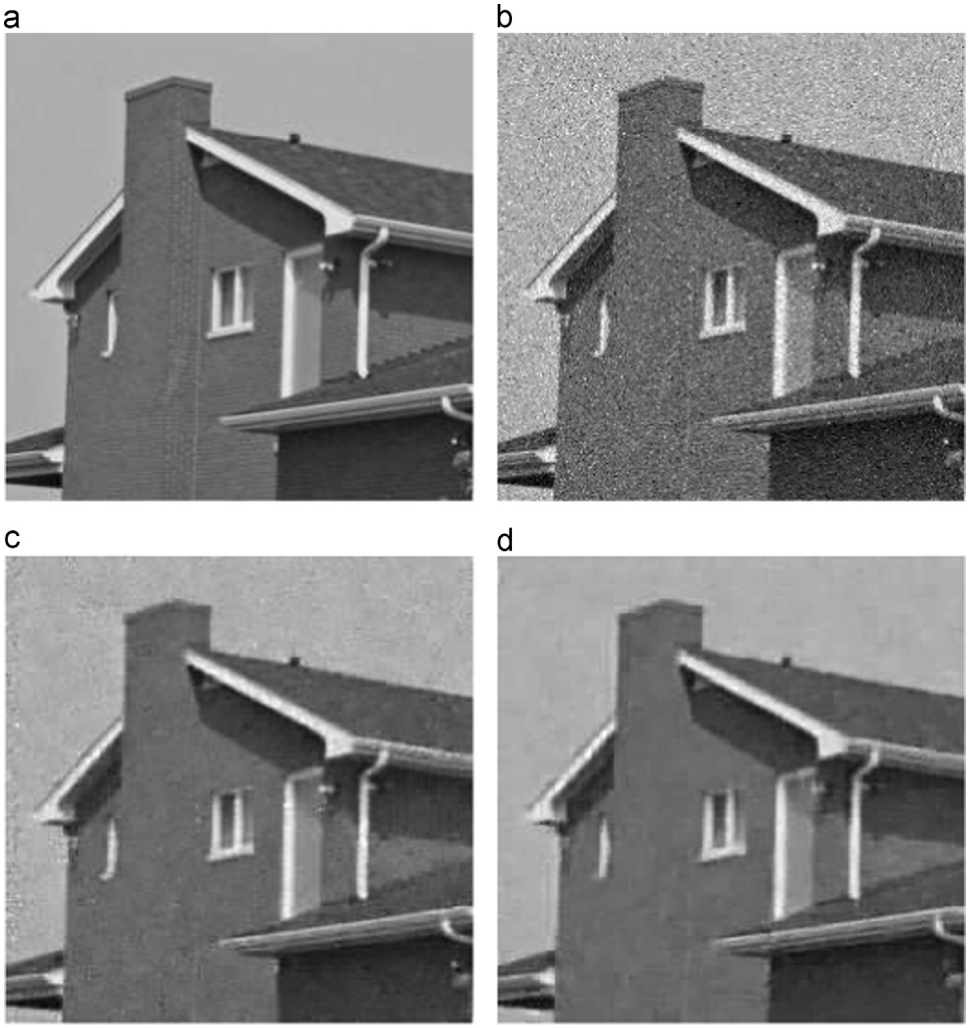


**Fig. 6.** Comparison of denoising performance of the K-SVD and $\ell_1$-K-SVD algorithms for Laplacian noise: (a) ground truth (clean) image; (b) noisy input, PSNR = 22.16 dB, SSIM = 0.36; (c) denoised image using K-SVD, PSNR = 29.78 dB, SSIM = 0.73; (d) denoised image using $\ell_1$-K-SVD, PSNR = 30.34 dB, SSIM = 0.81. The values of $\lambda$ and $n_p$ chosen in the experiment are 1 and 0.05, respectively.

perform well empirically. The value of $p$ is chosen to be $\frac{1}{2}$ in the experiment. A comparative analysis of the $\ell_p$-K-SVD and K-SVD is shown in Fig. 5, where the ADR and the distance metric $\kappa$, defined in Section 3.1, are plotted against the number of iterations. We observe that after 80 iterations, $\ell_p$-K-SVD recovers almost every atom (98%) of the ground truth dictionary accurately, whereas the ADR of K-SVD saturates at approximately 87%. In comparison with K-SVD, $\ell_p$-K-SVD also results in superior performance in terms of the distance $\kappa$ of the recovered dictionary from the ground truth. Fig. 5 indicates that $\ell_p$-K-SVD is more suitable than K-SVD for sparse noise.

### 3.3. Application to image denoising

We conduct image denoising experiments for additive Gaussian and Laplacian noise. The peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) of the denoised images produced by the K-SVD and the $\ell_1$-K-SVD algorithms are compared. In our experiments, dictionary learning is performed adaptively using patches of size $8 \times 8$ taken from the noisy image [5], with an inter-patch overlap of 4 pixels in both horizontal and vertical directions. For example, given a noisy image of size $256 \times 256$, we extract 3969 patches to train a dictionary of size $64 \times 128$. Both K-SVD and $\ell_1$-K-SVD algorithms are initialized with the noisy patches, and 10 iterations are executed [5]. In the sparse coding stage of the K-SVD algorithm, we solve the following optimization problem:

$$\min_{\mathbf{x}_n} \|\mathbf{x}_n\|_0 \quad \text{s.t.} \quad \left\|\mathbf{y}_n - \hat{D}\mathbf{x}_n\right\|_2 \le 1.15\sigma, \tag{13}$$

where $\sigma^2$ is the variance of the additive noise and $\hat{D}$ denotes the current estimate of the dictionary. The factor 1.15 is determined experimentally [5]. The $\ell_1$-K-SVD update of the coefficients is obtained by solving (3) for each noisy patch $\mathbf{y}_n$, followed by hard-thresholding. The values of $\lambda_n$ are chosen to be equal for all patches, and denoted by $\lambda$. Hard-thresholding is performed by retaining a fraction $n_p$ of $K = 128$ entries in each column of $X$. The values of $\lambda$ and $n_p$ depend on the type

and strength of the noise. As a rule of thumb, one should increase $\lambda$ and decrease $n_p$ as $\sigma$ increases. The optimal values of the two parameters $\lambda$ and $n_p$, for various noise levels $\sigma$ are obtained experimentally and reported in Table 1. The estimated clean patches are stitched together with averaging in the overlapping regions to generate the denoised image. The PSNR and SSIM of the denoised images, for additive Laplacian and Gaussian noise of different standard deviations $\sigma$, are reported in Tables 2 and 3, respectively.

We observe that the $\ell_1$-K-SVD algorithm yields an output PSNR comparable with the K-SVD, and results in a higher value of SSIM over K-SVD, especially when $\sigma$ is large. The improvement in SSIM for low values of input PSNR is about 0.1. For images corrupted by Gaussian noise with low values of $\sigma$, K-SVD results in a better denoising performance than $\ell_1$-K-SVD. However, as $\sigma$ increases, $\ell_1$-K-SVD preserves the image structure better, as reflected in the higher SSIM values. When the image is corrupted with Laplacian noise, $\ell_1$-K-SVD yields slightly better PSNR over K-SVD, but the improvement in SSIM is in the range of 0.08–0.10. For visual comparison, we show the denoised output images obtained using K-SVD and $\ell_1$-K-SVD in Fig. 6, where the input image is contaminated with Laplacian noise, at a PSNR of 22.16 dB. The PSNR and SSIM of the denoised images obtained using the two algorithms are indicated in Fig. 6. We observe that the gain in PSNR using $\ell_1$-K-SVD is slightly better than K-SVD, and the improvement in SSIM is approximately 0.1.

## 4. Conclusions

We have developed a robust dictionary learning algorithm, referred to as the $\ell_1$-K-SVD, for minimizing the $\ell_1$ error on the data term and demonstrated successful application to image denoising. The motivation behind using the $\ell_1$ metric as a measure of data fidelity is to achieve robustness to non-Gaussian noise and to alleviate the problem of over-smoothing texture and edges in images. The $\ell_1$-K-SVD algorithm updates the dictionary and the coefficients simultaneously, offering better flexibility and faster convergence compared with the RDL algorithm in [30]. The use of IRLS to minimize the $\ell_1$ objective retains the power and simplicity of $\ell_2$ minimization, as one can compute the solution in closed form at every iteration. Experiments on synthesized data indicate the superiority of the proposed algorithm over RDL, as well as its $\ell_2$-based counterparts, namely the MOD and K-SVD algorithms, in terms of the atom detection rate and the distance of the recovered dictionary from the ground truth. We have also demonstrated the superiority of the $\ell_1$-K-SVD algorithm for the case where the number of training examples is limited. Our formulation extends naturally to the case where an $\ell_p$ metric, $0 < p < 1$, is used on the data term and the regularizer to enforce sparsity in the presence of impulsive noise. The resulting $\ell_p$-K-SVD algorithm is shown to outperform K-SVD when the additive noise is indeed impulsive. Image denoising experiments demonstrated that the $\ell_1$-K-SVD algorithm results in output images with slightly higher PSNR values, especially at low input PSNRs, compared with the K-SVD algorithm. However, the improvement in SSIM is significant over K-SVD. We hope that the $\ell_1$-K-SVD algorithm will also turn out to be useful in other image processing applications such as super-resolution and inpainting.

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